

PETNet: A YOLO-based prior enhanced transformer network for aerial image detection

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ABSTRACT

Unmanned aerial vehicles (UAVs) have been applied to inspect in various scenarios due to their high efficiency, low cost, and excellent mobility. However, the objects in aerial images are much smaller and denser than general objects, causing it difficult for current object detection methods to achieve the expected results. To solve this issue, a prior enhanced Transformer network (PETNet) based on YOLO is proposed in this paper. Specifically, a novel prior enhanced Transformer (PET) module and a one-to-many feature fusion (OMFF) mechanism are proposed to embed into the network. Two additional detection heads are added to the shallow feature maps. In this work, PET is used to capture enhanced global information to improve the expressive ability of the network. The OMFF aims to fuse multi-type features to minimize the information loss of small objects. In addition, the added detection heads provide more possibility of detecting smaller-scale objects, and the extended multi-head parallel detection is more suitable for the multi-scale transformation of objects in aerial images. On the VisDrone-2021 and UAVDT databases, the proposed PETNet achieves state-of-the-art results with average precision (AP) of 35.3 and 21.5, respectively, which indicates that the proposed network is more suitable for aerial image detection and is of a great reference value.

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1. Introduction

Unmanned aerial vehicles (UAVs) have been widely applied in inspection, logistics, and prospecting [1,2], assisting with automatic object detection in aerial images [3]. Existing object detection methods are mainly divided into traditional manual feature extraction and deep learning-based [4]. In the recent decade, convolutional neural network (CNN) based methods have attracted attention due to their autonomous feature extraction and simple pipeline.

Current general object detection networks [5–8] exhibit outstanding performance in conventional scenarios while encountering challenges when applied to aerial images. This is because UAVs usually overlook from high space such that objects in aerial images are small and dense, as shown in Fig. 1 (a). On one side, the size of most objects is smaller than 32×32 pixels, which makes it easy to lose feature information [9], as shown in Fig. 1 (b). On the other side, the objects are relatively dense in streets, parks, and plazas. The dense distribution of objects also makes it

difficult for the network to extract effective features, resulting in many missed instances. In addition, during the flight of the UAV, the object scale varies drastically due to the change in altitude and viewing angle. The detection of multi-scale objects requires the network to have a robust fitting ability and scale invariance. All the above factors affect the accuracy and performance of the detection methods.

To address the above problems, many scholars have conducted related research. Li et al. proposed a density-map guided object detection network (DMNet) [10]. The model used a density map generation network to perceive object regions and combined an image cropping strategy to complete detection. Duan et al. constructed a coarse-grained density map network (CDMNet) [11], which generated coarse-grained density maps using the proposed lightweight dual-task density estimation network. Then the object regions were obtained by the cluster region generation algorithm. Li et al. stacked multiple convolutional layers with different convolution kernel sizes to extract multi-scale features. The proposed image cropping algorithm and outlier-aware non-maximum suppression were also effective [12]. Liang et al. formed an additional pyramid through the average pooling in the deconvolution module of FSSD and performed object detection by fusing two pyramid features. The spatial context was introduced to improve the detection

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Nomenclature

UAV	Unmanned aerial vehicle	CSPDark	CSP-ized DarkNet
CNN	Convolutional neural network	CSPSPP-PET	Constructed by CSPSPP and PET
DMNet	Density-map guided object detection network	CSPDark-PET	Constructed by CSPDark and PET
CDMNet	Coarse-grained density map network	ClusDet	Clustered detection
PETNet	Prior enhanced Transformer convolutional neural network	ESOM	Extreme scale objects metric
PET	Prior enhanced Transformer	DNOD	Dual neural network object detection
OMFF	One-to-many feature fusion	BSSD	Bag-based single-stage detector
TEM	Token enhanced module	PVT	Pyramid vision Transformer
PAM	Prior attention mechanism	FPN	Feature pyramid network
CSP	Cross stage partial	PAN	Path aggregation network
CSPSPP	CSP-ized spatial pyramid pooling	AP	Average precision

performance [13]. Although existing methods can effectively improve network performance, most of them mainly focus on one improvement strategy, such as density distribution or multi-scale feature fusion. The performance of the model could be further improved if considered comprehensively from multiple perspectives. Moreover, most existing methods are CNN-based, whose local convolution operation limits the ability to obtain global contextual information, resulting in the loss of spatial correlation [14]. Compared with CNN, the self-attention mechanism in the Transformer [15] can effectively extract global information, which is beneficial for the localization of object regions [16–18].

Based on the above points, we comprehensively consider the construction of the model from three aspects, including detection head, global density information, and multi-scale feature fusion. A prior enhanced Transformer convolutional neural network (PETNet) is proposed in this paper to meet the challenges of object detection in aerial images, including small-scale, dense instances, and multi-scale variations. By leveraging the current best general detection network Scaled YOLOv4 [19] as the baseline, PETNet embeds two additional detection heads, prior enhanced Transformer (PET) modules, and a one-to-many feature fusion (OMFF) mechanism. It is known that deep features (large receptive fields) are more conducive for detecting large objects, while shallow features (small receptive fields) are more suitable for small [20]. Therefore, two detection heads are added at the shallow layers to detect smaller-scale objects and improve the detection ability of multi-scale objects simultaneously. The enhanced global information is obtained by the PET, wherein the token enhanced module (TEM), which is an Attention mechanism, emphasizes necessary tokens (object regions) and weakens useless tokens (background regions), improving the dense prediction performance to a certain extent. The prior attention mechanism (PAM) transmits the attention matrix of the previous encoders as prior information to the subsequent encoders to further improve the performance of PET.

Meanwhile, to fully exploit the robust performance of PET, the CSP-ized (Cross stage partial) [21] spatial pyramid pooling (CSPSPP) and DarkNet (CSPDark) in the Scaled YOLOv4 are used to form CSPSPP-PET and CSPDark-PET with PETs. Moreover, the OMFF mechanism is used to fuse the features obtained by CSPDark-PET with multi-level features to enrich the amount of information, compensate for the loss of small object information, and improve multi-scale detection performance. The main contributions of this paper are as follows:

1. Two additional detection heads are added at the shallow layers to handle the prediction problem of smaller-scale and multi-scale objects.
2. A PET module, including TEM and PAM, is proposed to obtain enhanced global information, and PETs are introduced to CSPSPP and CSPDark to construct CSPSPP-PET and CSPDark-PET.
3. An OMFF mechanism is proposed to fuse the features obtained by CSPSPP-PET with multi-level features to enrich the information of detection heads.
4. The PETNet is evaluated on the VisDrone-2021 [22] and UAVDT [23] databases. Experimental results suggest that the proposed network has better performance in aerial image detection.

The rest of the paper is organized as follows: Related works are introduced in Section II. Section III describes our proposed method in detail. Section IV discusses our experimental results. Finally, the conclusions are presented in Section V.

2. Related work

Currently, general classical object detection networks have been divided into two categories: anchor-based and anchor-free. The difference is that the anchor-based method requires pre-set of the anchor box, while the anchor-free method does not. One-

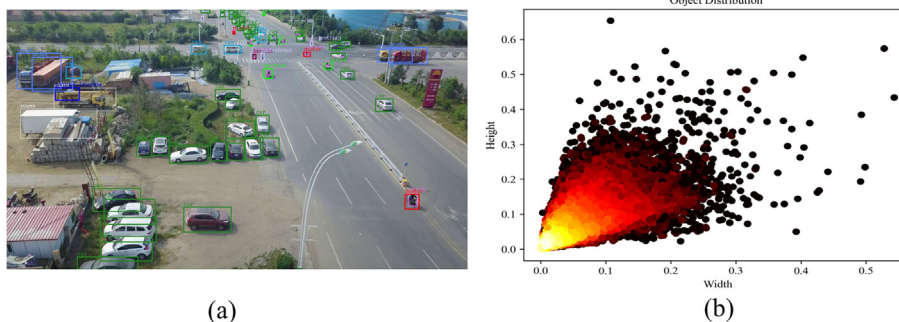


Fig. 1. An example image and the object scale distribution of the VisDrone-2021: (a) aerial image; (b) object scale distribution.

stage anchor-based methods represented by YOLOv3 [5] and RetinaNet [6] can directly perform regression and classification to the bounding box. The outputs of the network are regression parameters (anchor offsets) and category confidences. The mainstream two-stage anchor-based methods are Faster RCNN [7] and Mask RCNN [8]. Unlike the one-stage network, the two-stage network first extracts proposals, then followed by regression and classification of the bounding box. Similarly, numerous anchor-free methods have been proposed, such as CornerNet [24] and FSAF [25]. Among them, CornerNet is the classic of the keypoint detection network, and FSAF contains a feature selection anchor-free module to achieve anchor-free. Classic networks have achieved exciting results on natural images. However, they are limited by the characteristics of aerial images during detection.

In recent years, related CNN-based methods have been proposed to improve the detection performance of aerial images. The clustered detection (ClusDet) was proposed by Yang et al. [26], including three components: cluster proposal sub-network, scale estimation sub-network, and dedicated detection network. The network unified object clustering and detection in an end-to-end framework. The extreme scale objects metric (ESOM) network was constructed by Jin et al. [27]. The network first used metric learning to obtain scale representation boundaries for each object class, then merged indistinguishable small objects into small object regions and trained detectors on these regions. Tian et al. [28] proposed a dual neural network object detector (DNOD), which obtained comprehensive detection results by combining a one-stage and a two-stage detector for the judge and suspicious areas, respectively. Li et al. constructed an image pyramid detection model, which took the selected patch images and original images as the input of YOLOv3 and obtained detection results through an integrated decision-making algorithm [29]. Li et al. developed a bag-based single-stage detector (BSSD). The detector combined with the proposed adaptive bag definition method, positive sample definition method, and data augmentation algorithm to improve the effectiveness of the model further [30]. In addition, some studies focus on oriented detection algorithms. Oriented R-CNN, designed by Xie et al., included the oriented region proposal network for generating high-quality oriented proposals and the oriented R-CNN head for refining regions of interest and recognizing [31]. Cheng et al. proposed an anchor-free directed proposal generator. The coarse alignment module was applied to generate coarse-oriented boxes. The refined oriented boxes were passed through the RCNN head to produce the final detection results [32]. Furthermore, they constructed a double alignment oriented detector to solve the problem of spatial and feature misalignment [33]. Aiming at the boundary discontinuity and regression scheme inconsistency in the two-stage oriented network, Yao et al. proposed a bounding box representation method [34].

Most of the existing aerial image detection models are based on CNN. However, the local convolution calculation of CNN lacks in extracting global information. In contrast, the Transformer [15] captures the global information of entire feature maps by computing the correlation of feature patches, which can improve the perception ability of the model for large-scale complex scenes. Due to the significant performance of the Transformer, it has been emerging in the computer vision field since the rise of the Vision Transformer [16] in 2020, wherein a Transformer structure is used for image classification. Experiments suggest that its performance is better than the state-of-the-art CNN model. Successively, some works employ the Transformer to construct the backbone for feature extraction, including the Swin Transformer [17], Tokens-to-Token ViT [18], and pyramid vision Transformer (PVT) [35] consisting of the pure Transformer. Differently, CVT [36] and LeViT [37] consist of the CNN-embedded Transformer. However, the Transformer has disadvantages in computational complexity, and its

training time cannot be ignored. One intuitive way to reduce the complexity is to introduce the Transformer as a module into the CNN-based model so that the network can extract global information to improve the overall performance.

3. Prior Enhanced Transformer Convolutional Neural Network

Aerial image detection has always been challenging due to the characteristics of small-scale, dense instances, and multi-scale variations. In this paper, we propose PETNet to deal with the above challenges, with the structure shown in Fig. 2 (a). Specifically, the parameter number and the conventional image detection accuracy of YOLOv4-P5 are balanced compared to the other four models in Scaled YOLOv4, which is fair to verify the effectiveness of our proposed method. Therefore, the YOLOv4-P5 is used as the baseline. Then, two additional detection heads are embedded at shallow levels of YOLOv4-P5 to fully capture small object information. Meanwhile, the added detection heads provide the possibility to detect smaller-scale objects, which improves the fitting ability of the network and further enhances the scale invariance. Crucially, PETs are fused in the bottom and top layers of YOLOv4-P5. This module can make the network focus on the object region while ignoring the background region, which is effective for dense object detection. In addition, OMFF combined with the enhanced global information extracted by PET can achieve sufficient aggregation of multi-type features (e.g., global, local multi-scale, deep semantic, shallow spatial, multi-level information) to enrich the information of the network. This is straightforward to improve the detection performance of small or multi-scale objects.

As shown in Fig. 2 (a), the feature maps in the backbone and neck are annotated as B (backbone), F (feature pyramid network (FPN) [38]), and P (path aggregation network (PAN) [39]), respectively, and subscripts 1 ~ 5 correspond to the feature maps of different scales. Each module will be introduced in detail below.

Baseline. The detailed structure and explanations of the baseline are shown in Fig. 2 (b) [19]. In YOLOv4-P5, all modules such as DarkNet and SPP used by YOLOv4 [40] are CSP-ized and rebuilt as CSPDark and CSPSP to reduce computational costs while ensuring detection accuracy. The number of bottleneck layers (1×1 Conv, 3×3 Conv, and residual connection are shown in Fig. 2 (b)) L of each CSPDark in the backbone is 1-3-15-15-7, respectively, and the L of all CSPDarks in the neck is 3. Importantly, the CSPDarks at the neck do not use the residual connection [41]. Compared with PETNet, the feature fusion of YOLOv4-P5 is simpler, and there are only three relatively deep detection heads. Moreover, YOLOv4-P5 does not contain the extraction of global features.

Detection Head. Two additional detection heads are embedded on the P_1 and P_2 levels to process high-resolution shallow feature maps. Since the down-sampling of the feature map is applied, the information of the small object is lost while the receptive field is increased. Hence, deep features are more conducive for detecting large objects. On the contrary, the small object information is less lost in shallow layers, and detecting small objects does not require a large receptive field, so adding the P_1 and P_2 detection heads can significantly improve the detection ability of smaller-scale objects. Compared with the baseline of three detection heads (P_3, P_4 , and P_5), the parallel computation of five detection heads improves the fitting ability and scale invariance of objects with different scales, which is effective for multi-scale object detection. Moreover, as shown in Fig. 2 (a), we extend the neck structure, i.e., adding F_1, F_2, P_1 , and P_2 to accommodate the embedding of additional detection heads.

Prior Enhanced Transformer. As shown in Fig. 3, a complete PET pipeline consists of a front patch position embedding module and a PET module. The patch position embedding module is con-

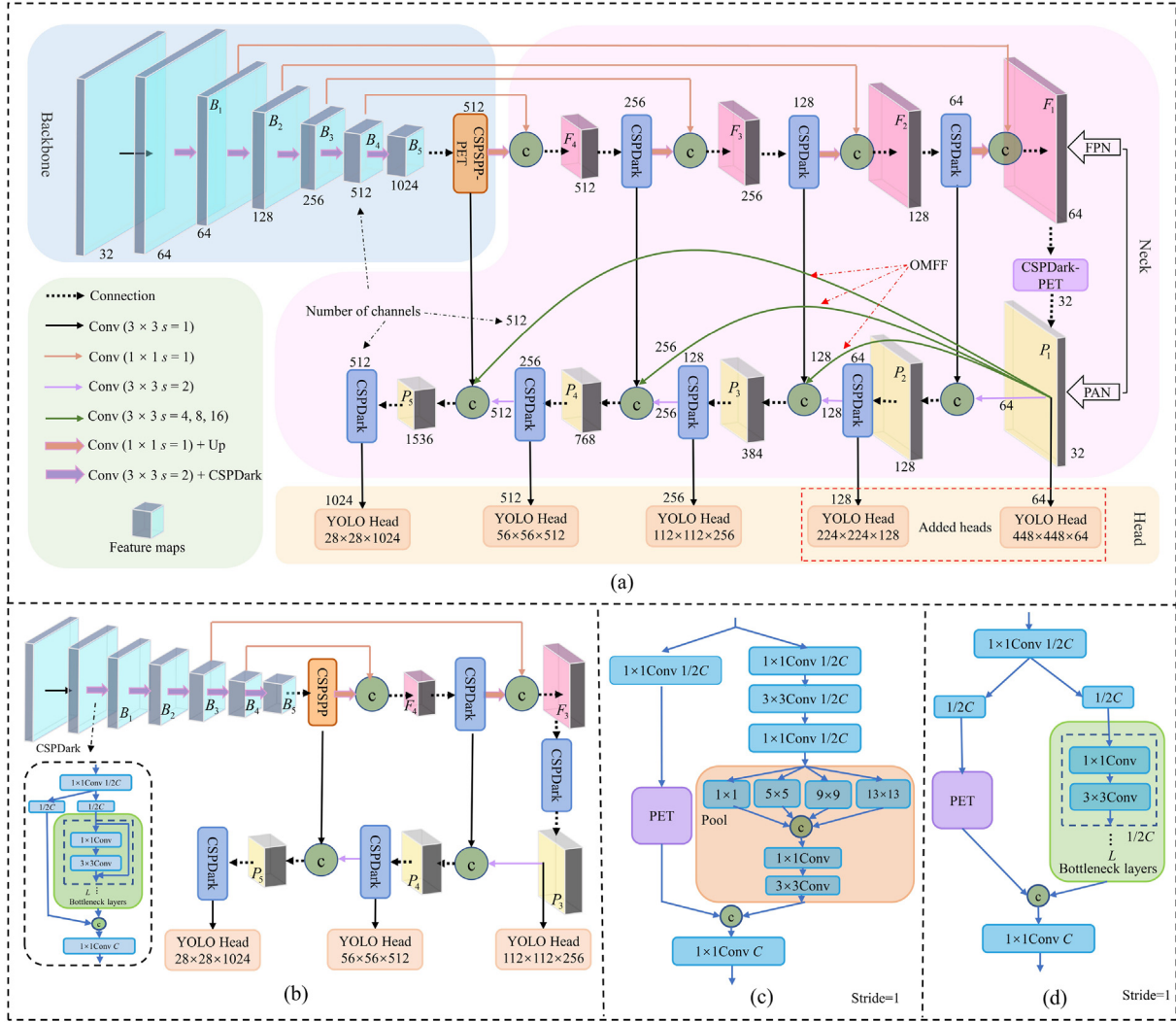


Fig. 2. Overall structure of PETNet: (a) PETNet; (b) YOLOv4-P5 (baseline); (c) CSPSP-PET; (d) CSPDark-PET, where Conv is constructed by convolution, batch normalization (BN) [42], and Mish activation function [43], s is the stride, Up is the up-sampling, Pool is the pooling layer, L is the number of bottleneck layers, C is the number of channels, and c is the concatenation.

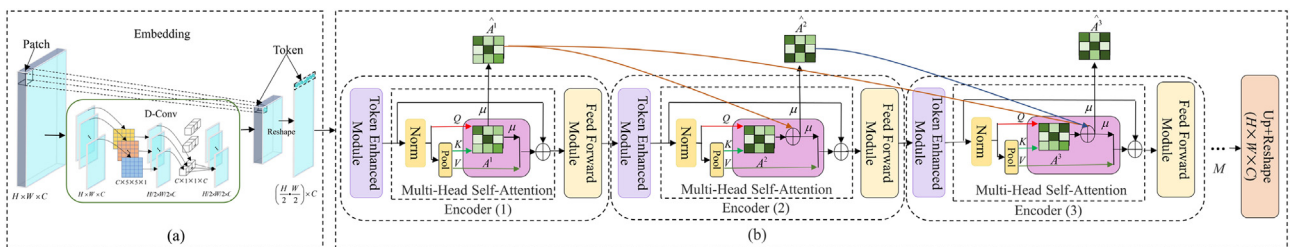


Fig. 3. Overall structure of the patch position embedding module and PET: (a) patch position embedding module; (b) PET, where C , H , and W are the number of channels, height, and width, respectively, D-Conv is the depthwise separable convolution, M is the number of encoders, μ is the Softmax, and Norm is the layer normalization.

structed by a depthwise separable convolution [44], including a depth-wise convolution and a point-wise convolution, to obtain tokens with positional embeddings, as shown in Fig. 3 (a). In this module, first, C convolutions with kernel $5 \times 5 \times 1$ and stride 2 are used to perform convolution calculations for each channel of the input feature map with the dimension of $H \times W \times C$ to obtain down-sampled output features, where C , H , and W are the number of channels, height, and width, respectively. Obviously, each pixel point in the output feature map contains 5×5 patch information.

In addition, the sliding of the convolution kernel endows the patches with a certain overlap, maintaining the correlation of the tokens. Then, C convolutions with kernel $1 \times 1 \times C$ and stride 1 are applied to fuse the feature map in the channel dimension, which is equivalent to the embedding process. Thus, the feature map of $H/2 \times W/2 \times C$ is acquired. To a certain extent, the zero-padding operation of convolution can be used for positional encoding [45], so no other position encoding method is introduced. Compared with ordinary convolution, depthwise separable convolution

reduces the parameter number effectively. Moreover, it also improves the ability to extract local information compared to dimensional transformation (nn.Unfold). Finally, the feature map is reshaped as $(H/2 \cdot W/2) \times C$ and forwarded into PET for global information extraction, where C is the embedding dimension (the number of channels) and $H/2 \cdot W/2$ is the number of tokens N .

The PET module is constructed by stacking M encoders. Fig. 3 (b) illustrates the detailed structure of the first three encoders. Each encoder contains a proposed TEM given in Fig. 4, a multi-head self-attention, and a feed forward module. As shown in Fig. 4, a fully connected layer with output dimension $C/2$ is used to reduce the dimension of the token vector, then the Mish [43] activation function is applied to add nonlinearity. The activated vector is fed into the second fully connected layer and the Sigmoid activation function to obtain the important weight of each token as follows:

$$v_n = \sigma(\delta(f_n W_1) W_2), n \in \{1, 2, \dots, N\} \quad (1)$$

where N is the number of tokens, $f_n \in \mathbb{R}^C$ is the n -th token vector, $W_1 \in \mathbb{R}^{C \times \frac{C}{2}}$ and $W_2 \in \mathbb{R}^{\frac{C}{2} \times 1}$ are the weights of the first and second fully connected layers, $\delta(\cdot)$ and $\sigma(\cdot)$ are the Mish and Sigmoid activation functions, and $v_n \in \mathbb{R}$ is the weight of the n -th token vector. Finally, the weight vector $v \in \mathbb{R}^N$ and the input feature matrix $F_t \in \mathbb{R}^{N \times C}$ are dot-multiplied to obtain the output matrix $O_t \in \mathbb{R}^{N \times C}$ of TEM as follows:

$$O_t = F_t \cdot v \quad (2)$$

Notably, the weight vector v can be obtained by learning, and each token corresponds to the 5×5 patch information in the original feature map. Consequently, the weight vector indirectly enhances the object regions and weakens the background regions, which can solve the problem of dense prediction to a certain extent.

In the multi-head self-attention of a single encoder, a pooling (AdaptiveAvgPool) layer is first applied to reduce the dimension of the output matrix of the layer normalization to obtain the reduced K (Key) and V (Value) matrices, referring to the PVT-V2 [46]. Reducing the matrix dimension can significantly reduce complexity and shorten training time. Then the full-dimensional Q (Query) matrix performs attention computation with K and V matrices [15]. The calculation process is shown in Eq. (3)~(6), where $W^Q, W^K, W^V \in \mathbb{R}^{C \times C}$ are the transformation matrices, $F_m \in \mathbb{R}^{N \times C}$ is the input of the multi-head self-attention, $O_{pool} \in \mathbb{R}^{(7 \times 7) \times C}$ is the reshaped output of the pooling layer, $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{C \times C_h}$ are the multi-head transformation matrices with C_h is the dimension of each head, h is the number of heads,

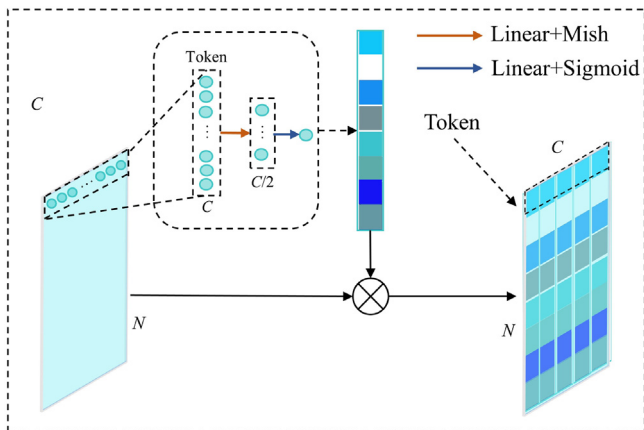


Fig. 4. TEM flow diagram, where C is the embedding dimension (the number of channels), N is the number of tokens, and Linear is the fully connected layer.

$C = h \times C_h, \hat{A}_i \in \mathbb{R}^{N \times (7 \times 7)}$ and $H_i \in \mathbb{R}^{N \times C_h}$ are the attention matrix and the output of the i -th head, $W^O \in \mathbb{R}^{h \times C \times C}$ is the output transformation matrix, $i \in \{1, 2, \dots, h\}$, $d_k = C$ is the dimension of the K matrix, and $O_m \in \mathbb{R}^{N \times C}$ is the output of the multi-head self-attention.

$$\begin{cases} Q = F_m \times W^Q \\ K = O_{pool} \times W^K \\ V = O_{pool} \times W^V \end{cases} \quad (3)$$

$$\hat{A}_i = \text{Softmax}(A_i) = \text{Softmax}\left(\frac{QW_i^Q \times (KW_i^K)^T}{\sqrt{d_k}}\right) \quad (4)$$

$$H_i = \hat{A}_i \times VW_i^V \quad (5)$$

$$O_m = \text{Concat}(H_1, H_2, \dots, H_i, \dots, H_h)W^O \quad (6)$$

Further, the output O_m is fed into the feed forward module to increase the generalization ability. We follow the design in PVT-V2 [46], and the feed forward module contains two fully connected layers and a depthwise separable convolution, wherein the depthwise separable convolution is used to add position information.

To further improve the performance of the Transformer, the proposed PAM is applied to PET. As shown in Fig. 3 (b), the attention matrix $\hat{A}$ of each encoder is added to the corresponding matrix A calculated by matrices Q and K of the subsequent encoder. For example, the attention matrix $\hat{A}^3$ of encoder 3 is obtained by adding the $\hat{A}^1$ in encoder 1, the $\hat{A}^2$ in encoder 2, and the A^3 obtained by its matrices Q and K . Therefore, the i -th head attention matrix of the j -th encoder is redefined as:

$$\begin{cases} \hat{A}_i^j = \text{Softmax}(A_i^j) & j = 1 \\ \hat{A}_i^j = \text{Softmax}(A_i^j + \hat{A}_i^{j-1} \dots \hat{A}_i^1) & j \in \{2, \dots, M\} \end{cases} \quad (7)$$

where M is the number of encoders, $i \in \{1, 2, \dots, h\}$, and h is the number of heads. The final output O_m^j of the j -th encoder is calculated by Eq. (5) and Eq. (6). In fact, the attention matrix $\hat{A}$ is transmitted sequentially and has a certain similarity to adjacent encoders. Hence, this paper uses an intuitive idea to regard the attention matrices $\hat{A}^{1 \sim (j-1)}$ of the previous encoders as prior knowledge and then densely transmits them to the subsequent encoder j to strengthen the expressive ability of the attention matrix $\hat{A}^j$ and further improve the overall performance of the Transformer. Finally, the output feature matrix of PET is up-sampled and reshaped to obtain a $H \times W \times C$ feature map that contains rich enhanced global information for subsequent convolution calculations. In this work, the number of encoders M and heads h are 3 and 8, respectively.

CSPSP-PET and CSPDark-PET. The PET modules are combined with the CSPSP and CSPDark modules in YOLOv4-P5 to form CSPSP-PET and CSPDark-PET. The structures are depicted in Figs. 2 (c) and (d). It can be seen that we embed the PET module into the short connected part of CSPSP and CSPDark and keep the others. The CSPSP-PET module is located at the deepest level (B_5) of the network and can effectively extract local multi-scale and global features of the deep semantic information. The CSPDark-PET module is located at the shallowest level (F_1) of the network, and its input features contain a lot of small object information from B_1 . Therefore, CSPDark-PET can effectively extract the global and local spatial information of small objects, making up for the shortcoming of the convolution's insufficient capture of the global context.

One-to-Many Feature Fusion. The OMFF mechanism is shown in Fig. 2 (a). First, four convolutions are applied to down-sample the P_1 feature map. Then, the down-sampled features are converted to concatenate with the outputs of CSPDarks and CSPSP-PET. In fact, the F_1 feature map contains part of the original B_1 shallow information. After global and local information extraction by CSPDark-PET, the P_1 feature map is used for feature fusion to enrich features. Therefore, the final output feature map obtained by OMFF consists of multiple types of features. Compared with the original Scaled YOLOv4, OMFF ensures that the output features of each level contain sufficient information to minimize the information loss of small objects. Meanwhile, multi-type (including multi-scale) feature fusion improves the robustness of the network, which is crucial for multi-scale object detection.

Under the above designs, PETNet significantly improves the detection performance of small and dense objects and reduces the impact of scale changes. The performance of PETNet is demonstrated through the experiment results below.

4. Results and Discussion

4.1. Databases

The VisDrone-2021 [22] database, provided by Tianjin University, is applied to test PETNet. Example images are shown in Fig. 5 (a). The database, captured by various drone-mounted cameras, contains more than 2.6 million objects from 14 cities. The VisDrone2021-DET is used for the object detection task. It consists of 6471 training images and 548 validation images in 10 categories, including pedestrian, person, bicycle, car, van, truck, tricycle, awning-tricycle, bus, and motor. The shooting environment covers various weather conditions, lighting conditions, and object density.

In 2018, Du et al. released the UAVDT [23] database at the European conference on computer vision. Fig. 5 (b) illustrates the example images. The database contains three categories of car, bus, and truck. In addition, 24143 training images and 16592 validation

images with an average resolution of 1024×540 are used for the object detection task. The database is collected from different locations in the city, such as squares, arterial streets, and toll stations. At the same time, it also covers different shooting conditions, including fog, rain, daylight, and night.

4.2. Implementation Details

All experiments are carried out on a server with GPU: three NVIDIA RTX 3090 (24G), memory: 64G, and CPU: Intel (R) Core (TM) i9-10920X. Mosaic [40], Mixup [47], and affine transformation are used to expand the training data. The MS COCO [48] average precision (AP), the area under the precision-recall curve, is applied to evaluate the proposed network. $AP_{IoU=0.50 \sim 0.95}$ is computed over ten different IoU thresholds (i.e., 0.50: 0.05: 0.95) and all categories.

In this work, we mainly use $AP_{IoU=0.50 \sim 0.95}$, $AP_{50}(IoU = 0.5)$, and $AP_{75}(IoU = 0.75)$. In addition, three additional metrics AP_S , AP_M , and AP_L representing small, medium, and large scales are also applied. The Adam [49] with an initial learning rate of 0.001, along with cosine annealing, is used to optimize the parameters. Meanwhile, other default tricks in Scaled YOLOv4, such as K-means automatic anchor selection, adaptive image scaling, mixed precision training, warm-up, and GloU Loss, are also used to optimize network training [19]. In the Visdrone-2021, the network is trained on the VisDrone2021-DET-train database, and the VisDrone2021-DET-val is used for evaluation. In the UAVDT, the 24143 training images are used to train the network, and the 16592 validation images are applied for evaluation. The input sizes are 896×896 and 640×640 for offline training and online detection, respectively.

4.3. Detection Results

The example detection results of PETNet on the VisDrone-2021 and UAVDT are shown in Figs. 6 (b) and (d), respectively. Figs. 6 (a) and (c) indicate ground truths. Fig. 7 shows the details of the detection results. As illustrated in Fig. 6, medium and large vehicle



Fig. 5. Example images of VisDrone-2021 and UAVDT: (a) VisDrone-2021; (b) UAVDT.

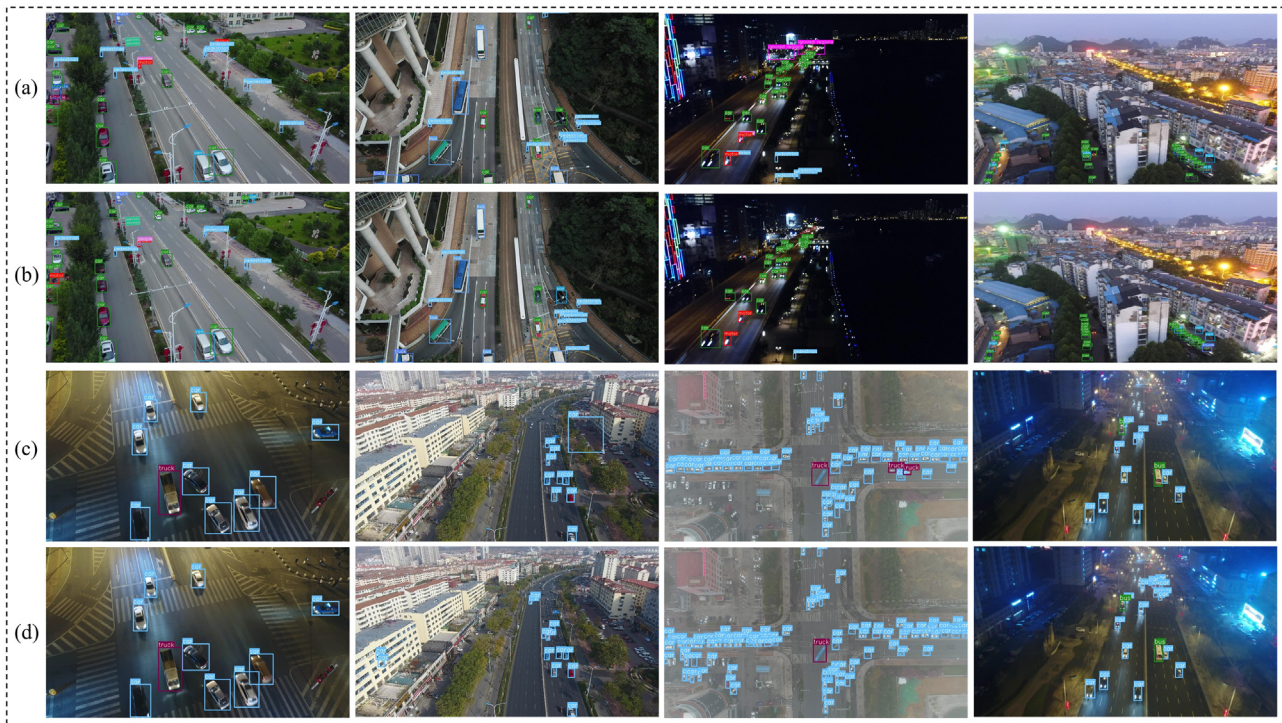


Fig. 6. Example detection results of PETNet on the VisDrone-2021 and UAVDT databases: (a) and (c) are ground truths of the VisDrone-2021 and UAVDT, respectively; (b) and (d) are the detection results of the VisDrone-2021 and UAVDT, respectively.

objects can be accurately identified and located by PETNet under different lighting and weather conditions. For small objects that can be distinguished by the naked eye, such as pedestrians and motors, PETNet can also obtain satisfactory results. Further, the detection results of challenging dense regions and multi-scale objects are competitive, as shown in the third column of Fig. 6. However, some false detection results exist in small objects with similar features, such as motors and bicycles, as shown in Fig. 7 (a). The similar appearance of different vehicles in the top view, such as vans, cars, and trucks, can also lead to false detection results, as depicted in Fig. 7 (b). However, these similarities affect negligible medium and large objects. Moreover, due to the high similarity of the objects, the database also has certain mislabeling. During the labeling process, there is no clear category boundary between some objects. The same class of objects can be labeled as different categories (e.g., pedestrians and people) in different images. These are the reasons for the false detection of the model. Figs. 7 (c) and (d) show the more common cases of missed detection. Most missed objects are too small to be distinguished by the naked eye at high resolution. In the case of poor lighting at night, the small object pixels are consistent with the background. Further, PETNet has some missed detection results in dense small object regions. The above problems are long-standing challenges for aerial image detection. Notably, PETNet can detect unlabeled occluded objects. As shown in Fig. 7 (a), the car with only one-third of the visible area can still be accurately located. Fig. 7 (b) verifies that PETNet can detect unannotated objects in the object-dense regions. In conclusion, the proposed PETNet has competitive detection results on both databases, demonstrating its superior performance in aerial image detection.

4.4. Ablation Experiments

4.4.1. PET

In this paper, the proposed TEM is used to emphasize necessary tokens (object regions) and weaken useless tokens (background

regions). The PAM transmits the prior information, the attention matrices of the previous encoders, to the subsequent encoders to improve the overall performance further. An ablation study is carried out on the VisDrone-2021 database to evaluate the effect of the two modules. The obtained results are shown in Table 1. From these results, the integration of PAM improves AP and AP_s by 0.5 and 0.2 points, respectively. When continuing to incorporate TEM, the AP and AP_s of PETNet can reach 35.3 and 27.0, which are 1.4 and 1.4 points higher than PETNet without both PAM and TEM, respectively. Meanwhile, other metrics have also improved to some extent. The ablation experiment results of the ten categories are shown in Table 2. When both modules are added to PETNet, the AP of all categories is effectively improved. The network performance improved by TEM is better than that of PAM. This is because TEM can make PET pay more attention to the object regions, and the object regions information is also available for the entire PETNet. While PAM only adds pseudo-prior information inside PET.

4.4.2. PETNet

The PETNet is constructed by the baseline, two additional detection heads, PETs, and OMFF. In this section, we verify the influence of different proposed modules. As shown in Table 1, the two additional detection heads can significantly improve the baseline performance. Compared with the baseline, the AP and AP_s are increased by 3.1 and 3.6 points, respectively. Further, by adding PET, the overall performance of the network is improved. In fact, the Transformer captures the dependencies between feature patches of the entire feature map through the self-attention mechanism, i.e., global information. For aerial images of large-scale and complex scenes, this can effectively improve the semantic recognition ability and associate scene information (e.g., relations between distant objects). However, convolution extracts local information within the kernel scale. Even if the receptive field is expanded, the range of information extracted is limited. In addition, TAM provides the Transformer with prior enhanced local attention infor-

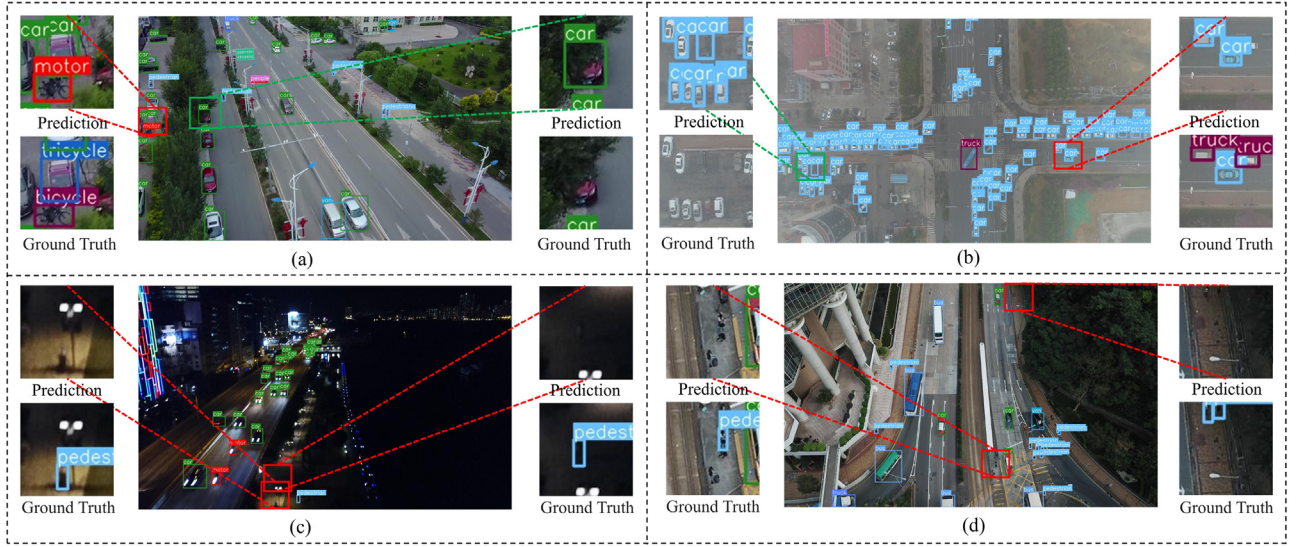


Fig. 7. Example detection result details. The red and green represent the error detection and the outstanding detection, respectively.

Table 1

The PET and PETNet ablation experiment results on the VisDrone-2021 database.

PET		PETNet				AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
PAM	TEM	Baseline	Heads	PET	OMFF						
✓	-	-	-	-	-	33.9	55.3	35.0	25.6	44.7	44.5
-	-	✓	-	-	-	34.4	55.6	35.8	25.8	45.6	46.3
-	-	✓	✓	-	-	30.5	50.8	30.8	21.3	42.8	45.6
-	-	✓	✓	✓	-	33.6	54.8	34.6	24.9	44.8	45.6
-	-	✓	✓	✓	✓	33.8	55.3	34.7	25.5	44.8	44.9
✓	✓	✓	✓	✓	✓	35.3	56.9	36.8	27.0	46.5	48.5

Table 2

The PET and PETNet ablation experiment results on ten categories of the VisDrone-2021 database (AP).

PET		PETNet				pedestrian	people	bicycle	car	van	truck	tricycle	awning tricycle	bus	motor
PAM	TEM	Baseline	Heads	PET	OMFF										
-	-	-	-	-	-	33.6	21.8	17.9	64.4	38.9	36.3	26.1	14.4	51.8	32.1
✓	-	-	-	-	-	33.6	22.0	19.3	64.7	38.7	36.3	27.2	14.1	54.0	32.2
-	-	✓	✓	-	-	28.2	18.2	15.4	59.5	35.8	33.5	23.6	12.4	48.9	27.9
-	-	✓	✓	-	-	33.5	21.4	18.1	64.4	39.1	35.6	24.3	14.8	50.8	31.7
-	-	✓	✓	✓	-	33.6	21.6	17.8	64.2	39.3	34.9	25.2	15.1	51.9	32.2
✓	✓	✓	✓	✓	✓	34.8	23.1	20.4	65.0	40.7	37.2	27.6	15.3	54.4	33.3

mation to emphasize necessary regions and weaken irrelevant regions. The use of PAM can further improve the overall performance of the Transformer. Therefore, embedding PET into the network can effectively improve detection performance. The PETNet obtains optimal performance when all proposed modules are integrated simultaneously. The AP , AP_{50} , and AP_{75} are 35.3, 56.9, and 36.8, respectively. Compared with the baseline, they are increased by 4.8, 6.1, and 6.0 points, respectively. In addition, the detection performance of the small objects improves the most compared to the other scales, and the AP_S is improved by 5.7 points. Combined with the additional scale information provided by the added detection heads, the enhanced global information obtained by PETs, and the original feature information of the baseline, the performance of OMFF is maximized. The fusion of multi-type features greatly enriches the information of the detection heads. Fig. 8 shows the detection results and heat maps of PETNet and baseline on the VisDrone-2021 database. The PETNet and baseline have comparable detection performance for large and apparent objects. But for small and dense objects, PETNet is more prominent. As shown in

heat maps, the perception area of PETNet is more accurate because that PETNet can better capture the effective feature points of a single instance. Table 2 shows the ablation experiment results on the ten categories. It can be seen that PETNet achieves optimal results in all categories, particularly for small objects such as pedestrians and people, and the AP increased by 6.6 and 4.9 points, respectively. Obviously, the proposed modules can effectively improve the aerial image detection performance of the network.

4.5. Comparisons with the classic networks

In this section, some classic general object detection networks, such as Cascade RCNN [50], FoveaBox [51], and SABL [52], are used to compare with PETNet on the VisDrone-2021 and UAVDT databases, as shown in Table 3 and Table 4. The detection results are depicted in Fig. 9 and Fig. 10. On the VisDrone-2021 database, PETNet achieves the best results, with AP , AP_{50} , and AP_S of 35.3, 56.9, and 27.0, respectively. Compared with the suboptimal Cascade RCNN, the AP and AP_S are increased by 12.1 and 13.2 points. More-



Fig. 8. Example detection results and heat maps on the VisDrone-2021 database: (a) PETNet; (b) baseline.

Table 3

Comparison results with classic networks on the VisDrone-2021 database.

Methods	Backbone	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Cascade RCNN [50] + FPN	ResNet101	23.2	38.4	24.1	13.8	35.0	42.4
CornerNet [24] + FPN	Hourglass104	19.0	30.2	19.8	14.4	27.8	16.0
FRCNN [7] + FPN	ResNet101	22.5	37.9	23.2	13.3	34.3	40.0
RetinaNet [6] + FPN	ResNet101	18.5	31.0	19.2	8.6	31.1	37.6
FSAF [25] + FPN	ResNet101	19.1	33.9	18.6	11.0	28.0	37.7
Fovea [51] + FPN	ResNet101	20.2	34.3	20.6	10.5	31.9	41.1
SABL [52] + RetinaNet + FPN	ResNet101	18.7	32.0	18.8	9.0	30.3	38.9
PETNet (Proposed)	CSPDarkNet-P5	35.3	56.9	36.8	27.0	46.5	48.5

Table 4

Comparison results with classic networks on the UAVDT database.

Methods	Backbone	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Cascade RCNN [50] + FPN	ResNet101	19.0	32.1	21.1	13.1	32.1	21.8
CornerNet [24] + FPN	Hourglass104	18.5	30.0	20.2	16.3	29.0	15.6
FRCNN [7] + FPN	ResNet101	19.0	31.9	21.1	13.2	32.1	22.4
RetinaNet [6] + FPN	ResNet101	21.1	34.7	23.7	14.6	34.5	20.9
FSAF [25] + FPN	ResNet101	19.8	35.6	20.0	13.3	32.0	20.3
Fovea [51] + FPN	ResNet101	21.3	36.5	23.2	14.3	33.9	28.9
SABL [52] + RetinaNet + FPN	ResNet101	19.9	36.4	19.8	13.7	32.4	16.4
PETNet (Proposed)	CSPDarkNet-P5	21.5	38.6	22.3	12.6	34.7	32.5

over, CornerNet acquires the highest AP_S of 14.4 except PETNet, and RetinaNet has the worst result of 8.6. Fig. 9 shows the detection results on the VisDrone-2021 database. For the classic objects, the detection results of all networks are satisfactory, but there are false detection results (e.g., cars and vans) in individual networks. The PETNet shows excellent detection results for small objects, followed by CornerNet and Cascade RCNN. On the UAVDT database, CornerNet and RetinaNet achieve the best AP_S and AP_{75} . However, their other metrics are relatively low. Notably, the AP_S of PETNet is not optimal because the data of UAVDT is severely imbalanced. As shown in Fig. 11, the UAVDT database contains three categories. The object number of “car” accounted for 93.31%. Moreover, most objects in the UAVDT database are small-scale. Compared with the VisDrone-2021 database, the object scale distribution is more uneven. The above problems cause missed and false detection results of PETNet. Furthermore, inaccurate and missing annotations in the UAVDT database further exacerbate the false detection. Fig. 10 indicates the detection results on the UAVDT database. The PETNet still has the most outstanding detection results. In contrast, the results of FSAF and Fovea are relatively poor. The parameter number and operating efficiency of the classic networks and PETNet are shown in Table 5, including floating point operations

(FLOPs), parameter number (Params), and frames per second (FPS). As indicated in Table 5, the Params, FLOPs, and FPS of PETNet are 83.03 M, 63.58 G, and 26.2, respectively. Obviously, the FLOPs and FPS are better.

4.6. Comparisons with the State-of-the-art

To further evaluate the performance of PETNet, some currently known networks dedicated to aerial image detection are used for comparison. The obtained results are shown in Table 6 and Table 7. On the VisDrone-2021 database, we can see that PETNet consistently outperforms the suboptimal network CDMNet by 3 ~ 4 points. Specifically, PETNet achieves the best results, with AP , AP_{50} , and AP_{75} of 35.3, 56.9, and 36.8, respectively. Compared with CDMNet, the AP , AP_{50} , and AP_{75} are improved by 3.4, 4, and 3.6 points, respectively. In addition, the AP_S is improved by 3.2 points, which indicates that PETNet effectively improves the detection performance of small objects. Table 7 shows the comparison results on the UAVDT database. Compared with the current networks, PETNet achieves optimal AP , AP_{50} , and AP_M of 21.5, 38.6, and 34.7, respectively. Although the AP_{75} , AP_S , and AP_L of PETNet are 22.3, 12.6, and 32.5, they are only 0.1, 1.3, and 2.7 points away

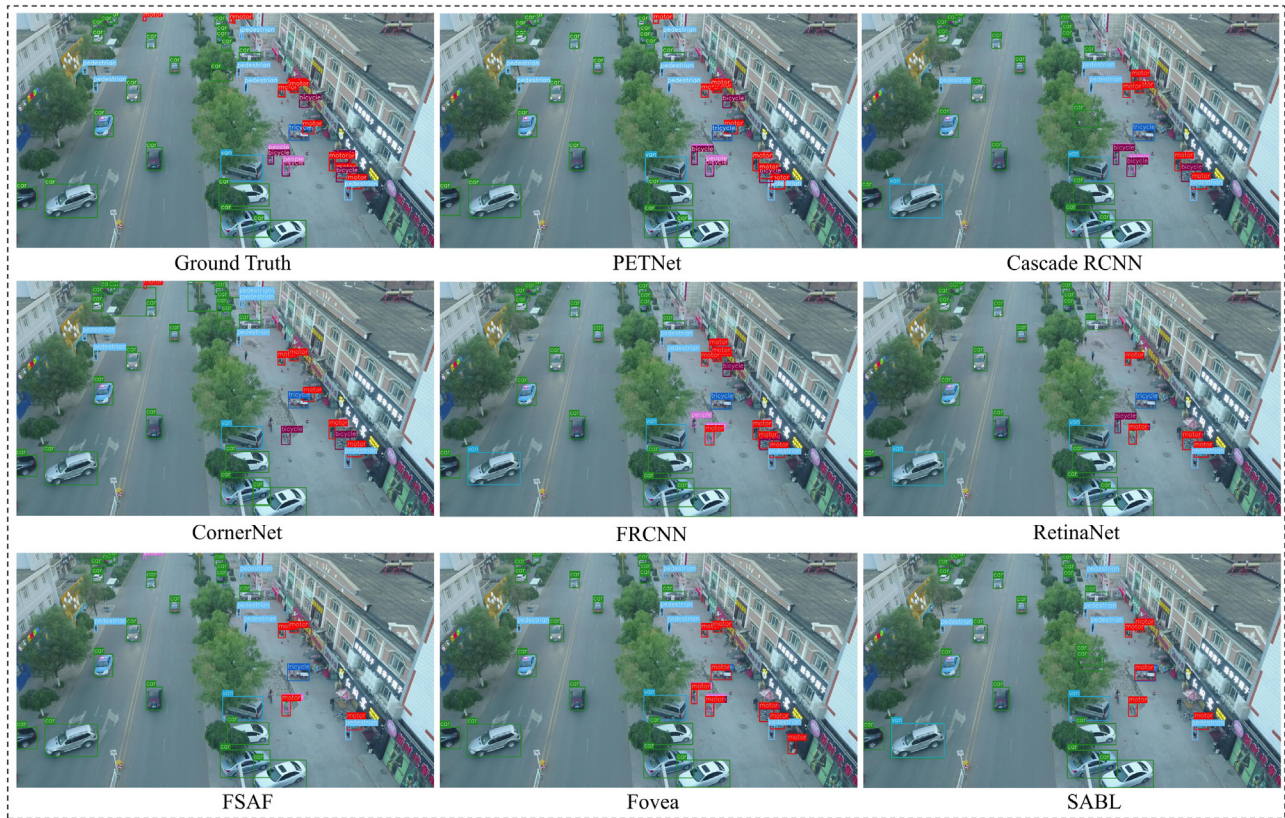


Fig. 9. Example detection results of the PETNet and classic networks on the VisDrone-2021 database.

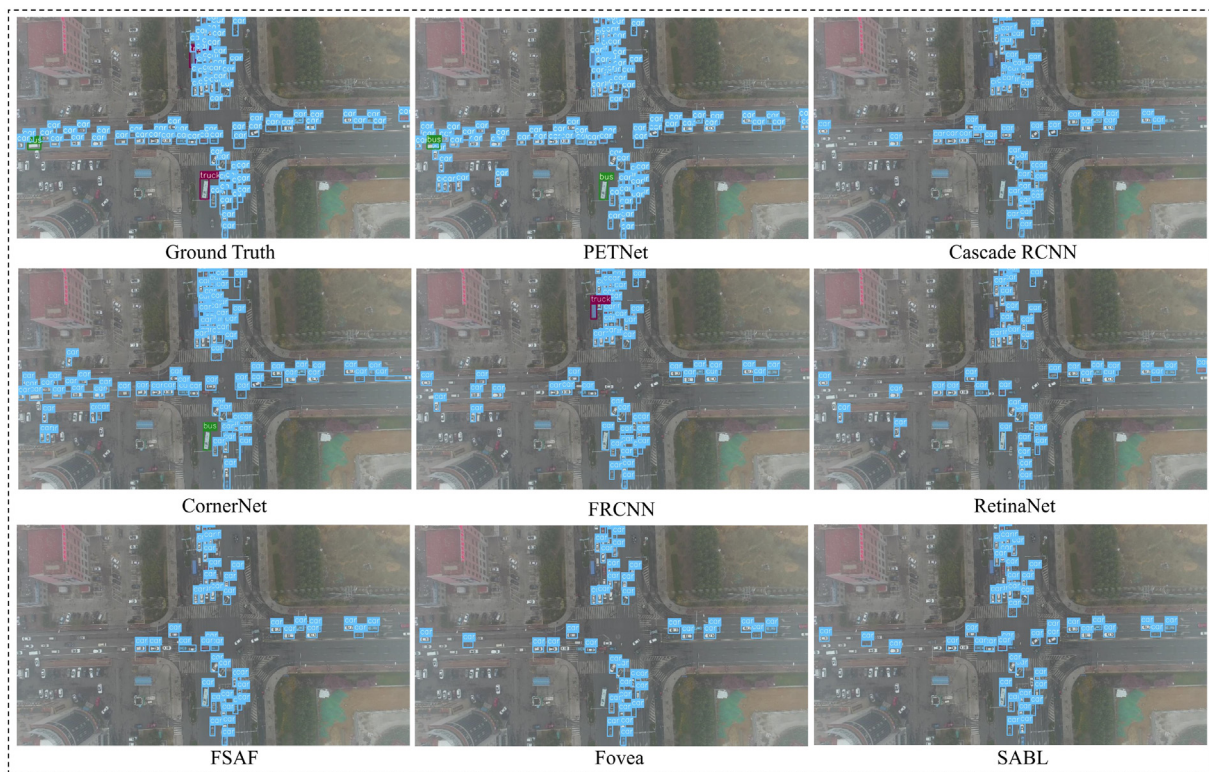


Fig. 10. Example detection results of the PETNet and classic networks on the UAVDT database.

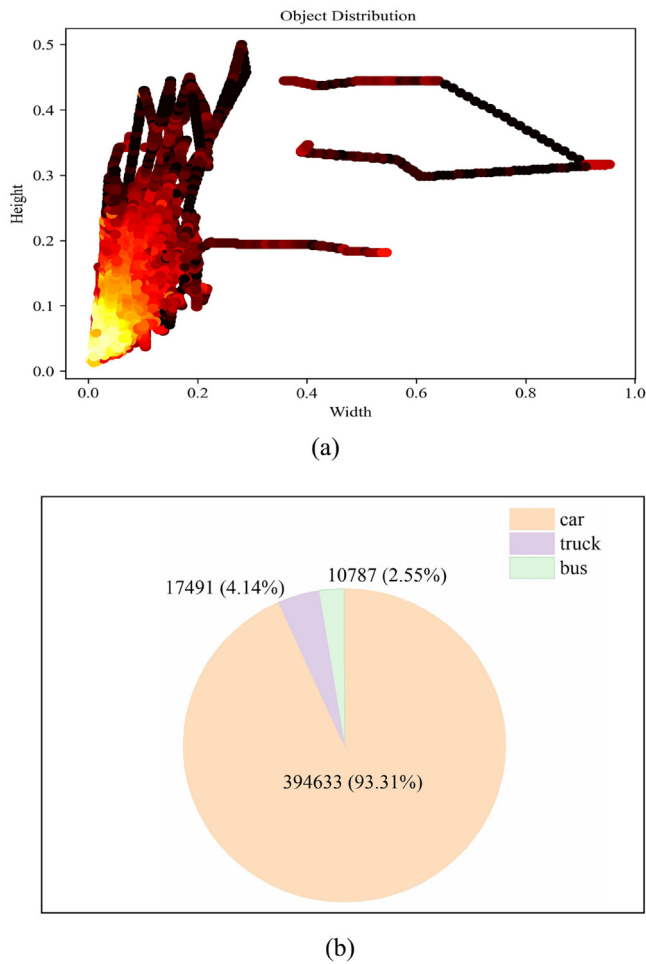


Fig. 11. Data distribution of UAVDT training data: (a) object scale distribution; (b) number of objects.

Table 5
Comparison results of the parameter number and operating efficiency.

Methods	Backbone	Pixels	Params (M)	FLOPs (G)	FPS
Cascade RCNN [50] + FPN	ResNet101	1333 × 800	87.95	285.35	14.9
CornerNet [24] + FPN	Hourglass104	511 × 511	200.96	451.66	4.2
FRCNN [7] + FPN	ResNet101	1333 × 800	60.16	283.34	17.1
RetinaNet [6] + FPN	ResNet101	1333 × 800	55.28	298.66	17.5
FSAF [25] + FPN	ResNet101	1333 × 800	55.03	292.87	17.5
Fovea [51] + FPN	ResNet101	1333 × 800	55.03	292.87	18.2
SABL [52] + RetinaNet + FPN	ResNet101	1333 × 800	55.15	295.56	16.4
PETNet (Proposed)	CSPDarkNet-P5	640 × 640	83.03	63.58	26.2

Table 6
Comparison results with current networks on the VisDrone-2021 database.

Methods	Backbone	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
ClusDet [26]	ResNeXt101	28.4	53.2	26.4	19.1	40.8	54.5
DMNet [10]	ResNeXt101	29.4	49.3	30.6	21.6	41.0	56.9
CDMNet [11]	ResNeXt101	31.9	52.9	33.2	23.8	43.4	45.1
PETNet (Proposed)	CSPDarkNet-P5	35.3	56.9	36.8	27.0	46.5	48.5

from the best results. In summary, the overall performance of PETNet on both databases is competitive.

5. Conclusion

In this paper, PETNet is proposed to deal with the challenges of aerial image detection. The PETNet composes the baseline, additional detection heads, PETs, and OMFF. Through ablation experiments, we evaluate the effect of TEM and PAM in the proposed PET on the network and further verify the potential of the proposed PET, OMFF, and two additional detection heads for network performance improvement. On both VisDrone-2021 and UAVDT databases, the PETNet achieves state-of-the-art results, which fully verify the effectiveness of the proposed network for aerial image detection.

In fact, there are still some limitations. The detection performance of large objects is still expected to improve. In addition, we also need to pay attention to the impact of data imbalance on the network. In the future, we will optimize PETNet for the above problems and consider constructing a more lightweight aerial image detection network.

CRediT authorship contribution statement

Tianyu Wang: Conceptualization, Methodology, Software, Writing - original draft, Data curation. **Zhongjing Ma:** Supervision, Writing - review & editing, Funding acquisition, Project administration. **Tao Yang:** Methodology, Investigation, Visualization. **Suli Zou:** Formal analysis, Writing - review & editing, Funding acquisition.

Data availability

The data used in the study is publicly available. The download link has been noted in the manuscript.

Table 7

Comparison results with current networks on the UAVDT database.

Methods	Backbone	AP	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L
ClusDet [26]	ResNet50	13.7	26.5	12.5	9.1	25.1	31.2
DMNet [10]	ResNet50	14.7	24.6	16.3	9.3	26.2	35.2
YOLOv4 [28] + DNOD	CSPDarkNet + VGG-19	14.2	31.9	11.0	10.8	24.9	31.8
EfficientDet-D7 [28] + DNOD	Efficient-B7 + VGG-19	12.9	32.0	10.9	10.1	25.6	29.2
ESOM-DZ [27]	ResNet50	14.1	27.4	13.2	9.3	26.0	32.3
CDMNet [11]	ResNet50	20.7	35.5	22.4	13.9	33.5	19.8
ISSON [12]	ResNet50	14.2	27.8	13.1	-	-	-
BSSD [30]	ResNet101	19.3	30.7	20.0	-	-	-
PETNet (Proposed)	CSPDarkNet-P5	21.5	38.6	22.3	12.6	34.7	32.5

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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